

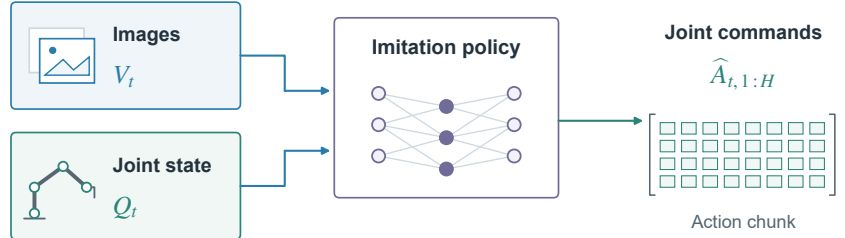
Blind Hands: Visual Reliance and Feedback Error in Dexterous Imitation

Anonymous submission

(a) Bottle-cap manipulation



(b) Which inputs guide the action?



Task success alone does not identify input use.

Fig. 1. Task and research question. (a) The robot positions its hands around a capped bottle. (b) Images and joint states are available to the imitation policy, but task completion alone does not identify how they influence commands. We diagnose input dependence and evaluate the effects of changing it.

Abstract—What information does an imitation-learning policy actually use to generate actions? Task success alone does not establish whether the policy uses visual scene information or exploits predictable joint-state correlations. We investigate this question in bimanual dexterous manipulation through visual and joint-state interventions, using tactile contact events to anchor exploratory diagnostics. We find weak image dependence in the diagnosed policy, including continued progression under frozen images in recorded-stream replay. Architectural and contact-aligned audits rule out explicit action-history copying and absent joint-state variation at contact onset. To address this visual underuse, we study action reparameterization, state noise, and dropout, asking whether encouraging visual dependence also improves command accuracy and feedback behavior. Experimental results show that, under full-state evaluation, 50% state dropout raises visual reliance from 0.392 to 0.990 relative to no dropout, while increasing command RMSE by 8.4% and left-hand numerical replay amplification from 2.09 to 15.97. By separating encoder state from command-composition state, our analysis identifies a feedback path retained when encoder state is absent and shows that exact relative targets preserve visual information conditional on state. Complementary noise and transformer comparisons reveal different response-error patterns. Together, these findings show why identifying input dependence and evaluating the consequences of changing it require joint measurements of visual response, command fidelity, and feedback error. Project page: https://blind-hand.github.io/blind_hands/.

I. INTRODUCTION

Learning from demonstrations [1], [2] spans image-to-control systems [3] and learned perception and control [4], [5]. Images describe the scene; proprioception describes robot configuration. Dense demonstrations also contain predictable state–command correlations. Task completion alone leaves unresolved which inputs drive an imitation policy.

Causal confusion [6], action copying [7], and uneven learning across task phases [8] expose related risks. Recent

studies report proprioceptive overdependence and poor spatial generalization [9], or limited visual contribution during motion transitions [10]. These findings motivate diagnosing input use before selecting an intervention. We call the weak visual response examined here *blind hands*.

We study policies based on implicit maximum likelihood estimation (IMLE) [11] and reproduce Action Chunking with Transformers (ACT) [12]. Visual/state interventions and tactile contact events anchor our initial diagnosis of an IMLE-based bottle-manipulation policy. The initial policy responds weakly to image changes and advances with frozen images in recorded-stream replay. Architecture and contact-aligned audits exclude explicit action-history copying and a strict lack of joint-state variation at contact onset. This diagnosis motivates attempts to strengthen visual guidance.

State conditions prediction, supplies the reference added to relative actions, and can support feedback correction. Our controlled dropout study finds greater visual reliance alongside higher command error and left-limb amplification. Consistency-noise comparisons and our ACT reproduction reveal different response–error patterns. Our analysis separates the retained composition path from learned state dependence.

We contribute a diagnosis of input dependence, a controlled counterexample to selection by visual reliance alone, and an analysis connecting command composition to feedback error.

II. RELATED WORK

A. Imitation Learning

Demonstrations enable dexterous control through reinforcement learning [13], human-video transfer [14], and arm–hand teleoperation [15], [16]. ACT, Mobile ALOHA,

and UMI advance bimanual learning and demonstration interfaces [12], [17], [18]. Multimodal action distributions are modeled with energy functions [19], discrete modes [20], diffusion [21], and IMLE generators [11]. We study IMLE and reproduce ACT, asking which observation channels drive these policies’ commands.

Sequential prediction also depends on feedback: a policy changes its future input distribution, allowing errors to compound [22]–[24]. DART collects corrective demonstrations by perturbing the supervisor [25]. Our noise comparison instead perturbs recorded state and recomputes relative targets, connecting to input-noise and dropout regularization [26], [27]. Following work on demonstration quality and evaluation variability [28]–[30], we fix the collection and training recipe and distinguish training-seed variation from episode uncertainty.

B. Causal Confusion

Causal confusion in imitation arises when a policy attributes expert actions to nuisance variables [6], a form of reliance on predictive shortcuts [31]. Copycat policies exploit past actions recoverable from observation histories [7]; keyframe-focused learning emphasizes action changepoints [8]. Our diagnosed policy has neither action-history input nor recurrent state, so this explicit copying path cannot explain its weak visual response. Current-state correlations remain a possible explanation.

Shared multimodal representations [32] can learn modalities unevenly [33], [34]. State-free Policy removes proprioception using relative end-effector actions and task-relevant visual coverage [9]; GAP attenuates proprioceptive gradients during motion transitions [10]. We examine relative joint commands and distinguish state used for prediction from state used to compose commands. Our interventions measure input dependence and its feedback costs; they do not identify causal confusion as the unique mechanism. The analysis in Section IV explains the retained feedback path and the information preserved by exact target subtraction.

III. DIAGNOSING WHAT THE POLICY USES

A. Input interventions and contact timing

The exploratory dataset contains $N_{\text{demo}} = 199$ bottle-cap demonstrations: 137,473 frames at 30Hz with joint states/actions, head RGB, and right-glove pressure. The generator predicts 16 commands from one observation, without action history or recurrent state. Visual/state substitutions fix other inputs; frozen-image replay tests progression without visual updates.

Tactile sensing provides a contact-time reference independent of the action residual. Joint-state variation persists across demonstrations at contact onset and after accounting for onset timing (Table I). This excludes a strict absence of variation at these events. Pressure locates contact; the comparison does not measure the benefit of tactile input. Appendix III specifies sensor quality and diagnostic scope.

TABLE I
CHECKS ON THE INITIAL POLICY AND EXPLORATORY COLLECTION.
EACH CHECK ADDRESSES A SPECIFIC EXPLANATION.

Question	Check and finding
Can the policy directly read its previous action?	The policy receives one observation, with no previous-action input or recurrent state.
Do joint states vary across demonstrations when contact begins?	Joint states vary across demonstrations at contact onset, including after onset-time regression.
Does the command sequence stop when images stop updating?	It continues to progress when replay repeats a fixed image.

B. From diagnosis to intervention

Weak image response and frozen-image progression are compatible with dependence on familiar movement regularities. They do not uniquely identify an internal representation or assign percentages of task success to inputs.

This diagnosis motivates interventions intended to increase visual use. Archived absolute/relative comparisons suggest larger image responses with relative targets, but their baseline checkpoint and training match are incomplete. The controlled dropout study therefore supplies the primary quantitative comparison; recovered noise and ACT pairs examine complementary settings. Each study retains its own evaluation scope.

IV. INTERVENTIONS, MEASUREMENTS, AND FEEDBACK

A. Conditional action model and two state paths

The robot has two 7-DoF arms and two 20-DoF hands. In the controlled study, the policy receives three RGB views V_t and joint state $Q_t \in \mathbb{R}^{54}$, and predicts a chunk of $H = 16$ commands. Each relative target is $D_{t,h} = A_{t,h} - Q_t$: every step subtracts the same current observation state. Figure 2 depicts this model and its evaluation paths. The earlier diagnostic and noise models use their recorded sensory configurations.

A hybrid vision transformer [35] encodes images. Its features and normalized state form the condition for an action transformer [36] with Gaussian latent Z . Inverse normalization yields relative predictions $f_\theta(V, Q_{\text{enc}}, Z)_h$ and composed commands

$$g_h(V, Q_{\text{enc}}; Q_{\text{act}}, Z) = Q_{\text{act}} + f_\theta(V, Q_{\text{enc}}, Z)_h. \quad (1)$$

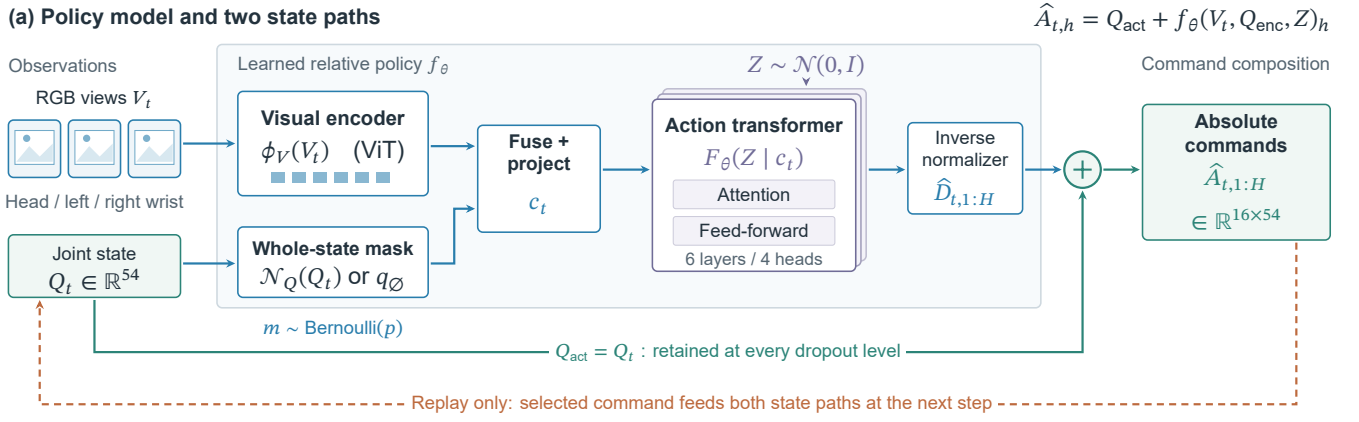
Encoder state conditions prediction; composition state converts the relative output to an absolute command. They coincide in ordinary inference. Fixing composition state during intervention isolates the learned response.

B. Whole-state dropout as a controlled intervention

For each training example, a Bernoulli mask $m \sim \text{Bernoulli}(p)$ replaces the entire normalized state vector with a learned token $q_\emptyset$:

$$q_{\text{enc}} = (1 - m)\mathcal{N}_Q(Q_t) + mq_\emptyset. \quad (2)$$

(a) Policy model and two state paths



(b) Joint evaluation of the composed command map g

Input dependence Swap V or Q_{enc} ; fix Q_{act}, Z . $R_{\text{vis}} = \frac{D_V}{D_V + D_Q}$ Also report absolute responses D_V, D_Q .	Command fidelity Predicted vs. demonstrated chunks $\text{RMSE}(S^{-1}(\hat{A} - A))$ Held-out windows; training-range scale S	Feedback propagation Recorded-state vs. predicted-state replay $G^{(\ell)} = \frac{\max_k E_{\text{auto}}^{(\ell)}}{\max_k E_{\text{teacher}}^{(\ell)}}$ Fixed $V, Z; \hat{Q}_{k+1} \leftarrow \hat{A}_{k,8}$
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Fig. 2. Architecture and evaluation of the controlled policy. (a) Three RGB views and masked encoder state condition a stochastic action transformer. The denormalized relative chunk is added to the retained composition state Q_{act} . The dashed loop feeds a selected command into both state paths during numerical replay. (b) Input interventions, demonstration error, and paired replay assess dependence, fidelity, and propagation, respectively. Paired replay uses the same recorded images and latent draws; it does not simulate environment responses.

Training uses $p \in \{0, 0.5, 1\}$ and a common IMLE-style objective [11], [37], [38]. Targets and composition state are unchanged, including at $p = 1$. Appendix I gives implementation details.

C. Three complementary measurements

Input response. For paired inputs, we substitute recorded images V^π or encoder state Q^π , retaining the recipient’s Q_{act} and latent draw. Let $S = \text{diag}(s_1, \dots, s_{54})$, where $s_j > 0$ is the maximum minus minimum training command for joint j , and $\bar{g}_h = S^{-1}g_h$. The empirical expectation averages equally over held-out episodes, selected windows, chunk steps h , and paired draws:

$$\begin{aligned} D_V^2 &= \mathbb{E} \|\bar{g}_h(V, Q; Q, Z) - \bar{g}_h(V^\pi, Q; Q, Z)\|^2 / 54, \\ D_Q^2 &= \mathbb{E} \|\bar{g}_h(V, Q; Q, Z) - \bar{g}_h(V, Q^\pi; Q, Z)\|^2 / 54, \\ R_{\text{vis}} &= D_V / (D_V + D_Q). \end{aligned} \quad (3)$$

$D_V, D_Q \geq 0$ are dimensionless root-mean-square (RMS) command changes. $R_{\text{vis}} \in [0, 1]$ is the image share of these responses; larger values indicate greater relative image sensitivity, not better task performance. Both components are needed to distinguish increased image response from decreased state response. We retain the archived zero/zero convention $R_{\text{vis}} = 0$ and omit it under missing-state input, where $D_Q = 0$ by construction.

Command fidelity. Let A_h be the demonstrated absolute command at chunk step h . The root-mean-square error on the same windows and draws is

$$\text{RMSE} = [\mathbb{E} \|S^{-1}(g_h - A_h)\|_2^2 / 54]^{1/2}. \quad (4)$$

This dimensionless error measures prediction fidelity; lower is better. Each metric is reduced within a trained model before averaging training-seed endpoints.

Feedback error. Two replay branches share recorded images and paired generator draws. The teacher-state branch uses recorded joint state; the autoregressive branch substitutes its preceding predicted command into both state paths. For branch $b \in \{\text{teacher}, \text{auto}\}$, let $\hat{Q}_{b,k+1}$ be its selected command and Q_{k+1}^* the next recorded state. With d_ℓ joints in limb ℓ ,

$$E_b^{(\ell)}(k) = \left[\frac{1}{d_\ell} \sum_{j \in \ell} w(\hat{Q}_{b,k+1,j} - Q_{k+1,j}^*)^2 \right]^{1/2}, \quad (5)$$

where $w(u) = ((u + 180) \bmod 360) - 180$ is the archived residual transform. Per episode and paired draw, amplification is

$$G^{(\ell)} = \frac{\max_k E_{\text{auto}}^{(\ell)}(k)}{\max_k E_{\text{teacher}}^{(\ell)}(k)}. \quad (6)$$

G is dimensionless: one means equal peak errors; values above one indicate amplification. Each model uses its own teacher-state denominator, which is nonzero in all evaluated cases. We report both peak magnitudes and G . The residual transform mixes recorded arm/hand units (Appendix II); its magnitudes are numerical scores, not physical-angle errors. Replay fixes images and excludes environment, servo, and contact responses.

D. What target subtraction changes

Exact relative targets preserve conditional visual information: $I(A - Q; V | Q) = I(A; V | Q)$. Matched functions

$f_D = f_A - Q$ also give identical commands and derivatives. This coordinate change alone does not determine learned feedback or accelerate visual learning (Appendix IV).

The retained feedback path. For a fixed image, latent draw, and chunk step h , the composed map with both state paths equal to Q has derivative

$$\frac{\partial g_h}{\partial Q} = I + \frac{\partial f_{\theta,h}}{\partial Q}. \quad (7)$$

A missing encoder token sets the learned state derivative to zero while retaining the identity term. More concretely, let Q_k^* denote recorded state and u_k the selected relative prediction under missing-state input. Under matched sensory inputs and latent draws, u_k is the same in both replay branches. With teacher residual $\eta_k = Q_k^* + u_k - Q_{k+1}^*$ and autoregressive residual $e_k = Q_k - Q_k^*$, initialization at $e_0 = 0$ gives

$$e_{k+1} = e_k + \eta_k, \quad e_k = \sum_{j < k} \eta_j. \quad (8)$$

Persistent residuals accumulate additively in ordinary coordinates, before the replay transform. Full-state models can supply learned correction. Donor sensitivity and the feedback derivative differ, motivating direct replay measurement; Appendix IV gives the bound.

V. CONSEQUENCES OF CHANGING INPUT DEPENDENCE

A. Experimental questions and controls

The controlled study uses a separate bottle-manipulation collection from six stores: head and bilateral wrist RGB, joint states/actions, and no tactile input. All dropout levels share 157 training and 42 held-out episodes and three training seeds. We evaluate stages 100/200 (rounded labels; Appendix I) with full encoder state (F) or the missing token (M). Q1 compares $p = 0.5$ versus $p = 0$ under F; Q2 compares $p = 1$ versus $p = 0.5$ under M. The remaining combinations are untrained-branch stress tests. Q3 tests consistency noise; Q4 tests absolute/relative targets in ACT. Their separate protocols delimit cross-setting conclusions.

Each held-out episode supplies three command-change windows and same-store donor observations. Replay selects command eight for 25 replays. Training data determine normalizers; the protocol fixes checkpoints, donors, and criteria before training (Appendix II).

We report individual seeds and 95% paired episode-bootstrap intervals conditional on the trained models and fixed donors. Leave-one-store-out analysis tests within-collection sensitivity. Figure 3 illustrates task outcomes; the quantitative comparisons below use recorded-data evaluation.

B. Q1: Does stronger visual use improve commands?

The full-state $p = 0.5$ versus $p = 0$ contrast isolates the effect of partial dropout. It increases visual reliance and absolute image response (Tables II and IV). At the final checkpoint, R_{vis} rises from 0.392 to 0.990. Image response D_V increases from 0.05963 to 0.10426, while state response D_Q decreases from 0.09240 to 0.00110. The difference from unrounded endpoints is 0.597 [0.571, 0.623].

RMSE rises from 0.05654 to 0.06127: +8.4%, ratio interval [1.054, 1.111]. Both stages satisfy the predeclared 10% point-estimate guardrail, but interval bounds exceed it, leaving noninferiority unestablished.

Left-arm and left-hand amplification rise from 2.25 to 4.04 and from 2.09 to 15.97. Both ratio intervals exclude one; right-limb intervals include it (Table VI). The left-arm/hand high-amplification fractions rise from 11.1%/11.1% to 87.3%/94.4%. These fractions count episodes whose inference-averaged $G \geq 3$, then average training seeds. Thus, the intervention strengthens image response but worsens command fidelity and left-limb feedback, failing the joint selection rule.

Absolute errors corroborate the increase: left-hand teacher-state peaks change from 0.300 to 0.305, while autoregressive peaks rise from 0.557 to 5.280. Peak magnitudes and episode-wise ratios have separate aggregation paths (Appendix II).

Repeatability. Reliance, command error, and left-limb amplification rise in every seed at both checkpoints; right-limb changes are mixed (Table VII). Checkpoints reuse the same trained models. Principal directions persist in all six store omissions. One early omission exceeds the 10% RMSE guardrail; the joint decision is unchanged.

C. Q2: Does removing encoder state improve accuracy?

Comparing $p = 1$ with $p = 0.5$ under M tests training-time encoder-state removal at fixed input availability. RMSE ratios are 0.999 [0.978, 1.024] and 0.993 [0.980, 1.008]. Both intervals admit higher or lower error, with two final-checkpoint seeds favoring partial dropout. This removal establishes neither accuracy superiority nor equivalence; left-hand amplification remains high.

D. Q3: Is the dropout trade-off inevitable?

An existing consistency-noise pair tests whether another intervention yields the same response-error pattern. This pair and ACT use the earlier collection. Their ratio $\rho_{\text{vis}} = \tilde{D}_V / \tilde{D}_Q$ divides mean target-output RMS responses to image and state substitutions. Its unbounded scale differs from R_{vis} ; comparisons remain within protocol (Appendix III).

In the existing one-seed comparison on two training episodes, consistency noise reduces both amplification and autoregressive peak error in all eight limb-episode combinations (Table III). The image/state response ratio rises from 0.576 to 1.534. Teacher-state errors vary across limbs, including a right-hand increase from 0.059 to 0.152 on one episode.

The pair combines state noise $\tilde{Q} = Q + \xi$ with recomputed targets $\tilde{D}_h = A_h - \tilde{Q}$, preserving $\tilde{Q} + \tilde{D}_h = A_h$. This joint intervention provides a different response-error pattern from dropout; the separate effects of noise and target recomputation, and broader generalization, remain unmeasured.

TABLE II

Q1: EFFECT OF 50% VERSUS NO STATE DROPOUT UNDER FULL-STATE EVALUATION. ESTIMATES AGGREGATE THREE TRAINING SEEDS; INTERVALS ARE CONDITIONAL 95% PAIRED EPISODE-BOOTSTRAP INTERVALS.

Measure	Unit	Stage 100		Stage 200	
		Change	95% interval	Change	95% interval
Visual reliance (R_{vis})	pp	+57.1	[+54.7, +59.2]	+59.7	[+57.1, +62.3]
Command RMSE	%	+8.8	[+5.7, +12.4]	+8.4	[+5.4, +11.1]
Replay amplification (G)					
Left arm	%	+63.1	[+50.4, +76.8]	+79.7	[+64.5, +97.7]
Right arm	%	+7.1	[−1.3, +18.8]	+7.0	[−3.7, +16.7]
Left hand	%	+572.1	[+502.4, +619.3]	+663.6	[+397.2, +728.6]
Right hand	%	−5.0	[−9.7, +10.7]	+4.2	[−4.6, +17.1]

pp: percentage-point difference; %: relative change from no dropout. Positive values indicate increases; zero means no change. Seed-level results appear in Table VII.

TABLE III

EXISTING CONSISTENCY-NOISE PAIR. EACH PAIR GIVES AMPLIFICATION FOR TRAINING EPISODES 0 / 60, IN THAT ORDER. ONE TRAINING SEED; NO POPULATION INTERVAL.

Limb	Without noise	Consistent noise
Left arm	3.79 / 2.63	0.93 / 1.01
Right arm	4.71 / 2.39	0.84 / 1.01
Left hand	2.46 / 2.42	1.08 / 1.03
Right hand	8.39 / 2.62	0.93 / 1.00

E. Q4: Do relative targets consistently improve visual use?

We reproduce ACT to compare absolute and relative targets. The one-seed ACT comparison shows substantial image response with absolute targets: at chunk length 100, ρ_{vis} is 2.04 and 2.35 across the two checkpoints. Relative targets fall below the historical fivefold visual-revival criterion. Replay comparisons favor absolute targets in 8/8 cells at chunk length 16 and 4/8 at length 100, across checkpoint and temporal-ensembling settings. These repeated measurements demonstrate that absolute-target policies can already be image-responsive and that the measured effect of target subtraction varies with architecture and configuration.

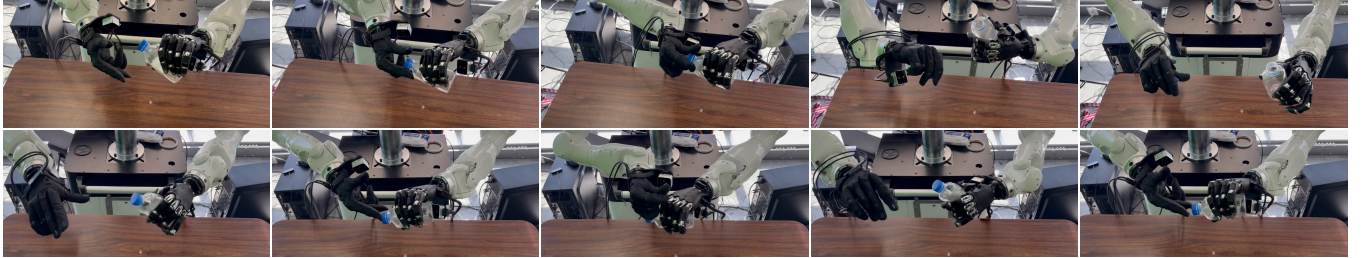


Fig. 3. Two recorded task sequences: cap removal (top) and cap still attached (bottom). Frames progress from left to right. The examples illustrate visible outcomes; they do not assign policy conditions or estimate method-level success rates.

VI. DISCUSSION AND CONCLUSION

Why diagnose blindness? Success in familiar conditions leaves unresolved whether commands follow the scene or joint-state correlations. A displaced object may require a different action at the same posture. Blindness must be assessed against task-relevant changes and execution phase; invariance to irrelevant imagery can be desirable. Our scores diagnose input dependence; appropriate correction also requires evaluating response direction, timing, and consequence.

Implications for hierarchical control. Hi Robot separates high-level reasoning from low-level action [39]; GR00T N1 couples a vision-language module to a motor-action generator [40]. DuoCore-FS combines asynchronous semantic updates with fast visual/proprioceptive control [41]. Our find-

ings suggest a criterion for this functional “big-brain/small-brain” division: preserve state feedback for local correction while checking that scene changes and updated subgoals influence commands. Accurate scene interpretation upstream need not influence the action policy. Separate visual and subgoal interventions could locate this loss of influence. For bottle opening, cap separation should switch the subgoal from twisting to lifting while local feedback regulates motion. These are design implications; hierarchical policies remain untested here.

A human-action analogy. Human reaching can adjust to a displaced visual target without conscious awareness of its displacement [42]. Automatic execution can thus retain visual feedback. This motivates testing scene-dependent cor-

rection alongside familiar-sequence completion. The analogy concerns functional coordination, not cerebral or cerebellar anatomy. Our dropout results show why encouraging visual response must also account for command fidelity and feedback error.

Scope and conclusion. Historical diagnostic provenance is incomplete; donor substitutions can be off-distribution, windows emphasize transitions, and bootstrap intervals condition on trained models and donors. Noise and ACT comparisons use one training seed. Replay fixes external observations and measures numerical error without contact or servo dynamics. Diagnose input use first, then assess visual response, command fidelity, and feedback error jointly.

APPENDIX I

IMPLEMENTATION AND REPRODUCIBILITY

Training. All nine primary models use one observation, three 224×224 RGB views, a ViT-small R26/S32 encoder, a six-layer/four-head action transformer, and 16-step chunks. Training uses batch size 32, 200 epochs, AdamW (learning rate and weight decay 10^{-4} ; betas 0.95/0.999), cosine decay, 1,000 warmup steps, and crop ratio 0.95. The visual encoder is trained throughout; EMA and consistency noise are disabled. Seeds 42/43/44 share split seed 42. The designated midpoint and final snapshots use nominal stage labels 100/200 rather than exact completed-epoch counts.

For target i and 20 Gaussian candidates j , let d_{ij} be the Euclidean distance between flattened normalized targets and candidates, and $d_{\max}$ its batch maximum. Training minimizes the mean of $b_i = \min_j \{d_{ij} + d_{\max} \mathbf{1}[d_{ij} \leq 0.03]\}$ over $b_i < d_{\max}$, or zero for an empty set. Relative subtraction precedes affine normalization fitted to absolute training commands. Inference uses one Gaussian latent per paired draw without candidate-selection history.

APPENDIX II

EVALUATION AND COMPLETE NUMERICAL RESULTS

Each held-out episode supplies three windows ranked by $\sum_j |A_{t+1,j} - A_{t,j}|$, separated by more than eight frames. This mixed-unit score can favor arm motion. Donors cycle within each held-out store, matching episode-progress deciles; camera views are swapped jointly. Inference seeds are 12345/999. Replay starts at frame five, selects command eight, advances eight frames, and resets each seed before every call for 25 replays.

Equation (5) applies the archived wrap before limb RMS. Software uses arm degrees and hand radians, with incomplete collection-time unit/version binding; w is therefore a numerical transform. An ordinary-residual check limited to six earlier seed-42/100-stage captures leaves hand scores and $G \geq 3$ fractions unchanged, but changes six arm episode-cell ratios.

Input-response and RMSE summaries average per-model endpoints across training seeds. Replay aggregation is defined below. The bootstrap jointly resamples episode indices across models and training seeds (2,000 draws; seed 20260905). The frozen selection rule requires a reliance

gain at both checkpoints, a final difference interval above zero, RMSE point ratios at most 1.10, no limb median- G ratio above 1.25, and no high-amplification fraction increase above 10 percentage points. These operational criteria and all 36 cells' endpoints and paired intervals are independently reproduced from archived arrays.

Table IV includes untrained branches as distribution-shift stress tests.

Table V summarizes episode maxima; dividing its columns does not reconstruct G , which aggregates episode-wise ratios. Command RMSE uses demonstrated commands; replay uses future recorded states, which may differ because of tracking lag.

To make the order of reductions explicit, index training seeds by s , episodes by i , and inference draws by r . For each limb, the reported endpoint is

$$\hat{G}_p = \frac{1}{3} \sum_s \text{median}_i \left[\frac{1}{2} \sum_{r=1}^2 G_{p,s,i,r} \right]. \quad (9)$$

Contrasts use unrounded aggregate endpoints, not per-seed ratios. Bootstrap replicates repeat the full reduction with shared episode indices and fixed donors. Intervals are point-wise across limbs.

Table VI gives final limbwise intervals; under M, only the right-hand interval excludes one.

APPENDIX III

EXPLORATORY AND COMPLEMENTARY PROTOCOLS

Exploratory diagnostics. Frozen-image replay repeats one image while numerical state evolves. Contact-aligned records compare joint states across demonstrations before and after onset-time regression. Incomplete checkpoint, split, and raw-array provenance limits quantitative reuse of these historical probes.

The right glove contains 24×32 taxels, including one stuck channel and 219 near-dead channels (28.5%). Contact timing uses masked robust pressure aggregation. The left hand has no tactile sensor. We use these events as a timing reference, not as a tactile ablation or a bilateral correlation measure.

Recovered consistency-noise pair. The noise pair uses the exploratory collection (Section III) with head RGB and right-glove pressure, sharing training seed, augmentation, batch size, and normalizers. It uses checkpoint 100 and training episodes 0/60; the noise-sampler revision is not bound to the checkpoint. The probe averages target-chunk RMS changes over 120 training frames (image donor 105388, state offset 40, fixed pressure), then divides image by state response. Paired inference seeds are 12345/999; replay uses ordinary differences and seed 12345.

ACT comparison. ACT uses head RGB and state, one observation, chunk lengths 16/100, the LeRobot 0.4.4 CVAE, 100,000 steps, batch size eight, seed 12345, and a 179/20 split. Relative targets subtract the observation state from every step; target-specific normalizers differ. Evaluation uses latent mean zero, training-frame probes, midpoint/final checkpoints, and both ensembling settings. Its raw training losses are not a matched-fit comparison.

TABLE IV

COMPLETE DESIGN: 36 CELLS REDUCED TO 12 MEANS OF THREE TRAINING-SEED ENDPOINTS. F/M: FULL/MISSING STATE; *: UNTRAINED BRANCH. G IS THE MEAN OF THREE EPISODE MEDIANS. LA/RA/LH/RH: LEFT/RIGHT ARM/HAND. MISSING-STATE RELIANCE IS OMITTED.

Stage	p	Mask	R_{vis}	RMSE	G_{LA}	G_{RA}	G_{LH}	G_{RH}
100	0	F	0.417	0.05838	2.45	2.20	2.34	1.89
100	0	M*	–	0.13022	5.15	5.02	16.52	3.76
100	0.5	F	0.988	0.06354	3.99	2.36	15.71	1.80
100	0.5	M	–	0.06374	4.02	2.34	16.01	1.80
100	1	F*	0.999	0.06367	4.18	2.54	16.18	1.71
100	1	M	–	0.06367	4.18	2.54	16.18	1.70
200	0	F	0.392	0.05654	2.25	2.19	2.09	1.68
200	0	M*	–	0.14456	4.71	5.30	16.05	3.64
200	0.5	F	0.990	0.06127	4.04	2.34	15.97	1.76
200	0.5	M	–	0.06147	4.03	2.34	16.12	1.75
200	1	F*	0.999	0.06106	3.95	2.31	16.04	1.64
200	1	M	–	0.06106	3.96	2.31	16.04	1.64

TABLE V

STAGE-200 PEAK REPLAY ERRORS. EACH COLUMN AVERAGES INFERENCE-DRAW MAXIMA, TAKES EPISODE MEDIANS, THEN AVERAGES TRAINING SEEDS. TEACHER/AUTO: RECORDED-STATE/AUTOREGRESSIVE INPUT.

p / Mask	Limb	Teacher	Auto
0 / F	LA	8.672	18.617
	RA	8.578	17.317
	LH	0.300	0.557
	RH	0.131	0.204
0.5 / F	LA	8.786	34.679
	RA	8.729	18.489
	LH	0.305	5.280
	RH	0.137	0.224
0.5 / M	LA	8.811	34.301
	RA	8.728	18.288
	LH	0.306	5.355
	RH	0.138	0.224
1 / M	LA	8.642	35.544
	RA	8.464	17.171
	LH	0.303	5.332
	RH	0.143	0.221

For these historical probes, $\tilde{D}_X = \mathbb{E}[\|f_\tau(o) - f_\tau(o^{\pi^x})\|_F / \sqrt{Hd}]$, where f_τ is the denormalized target-space output, $X \in \{V, Q\}$, and d is its action dimension. Within each inference seed, averaging over frames precedes the ratio; noise-pair summaries then average the two seed ratios. Target coordinates and donors remain protocol-specific. Historical cutoffs define weak response as $\rho_{\text{vis}} < 0.3$ and revival as a fivefold relative/absolute increase. These cutoffs do not apply to R_{vis} .

APPENDIX IV ANALYTICAL DETAILS

Missing-state recurrence. The triangle inequality applied to Eq. (8) gives

$$\max_{1 \leq k \leq T} \|e_k\|_\ell \leq T \max_{0 \leq j < T} \|\eta_j\|_\ell. \quad (10)$$

TABLE VI

FINAL-CHECKPOINT AMPLIFICATION CONTRASTS WITH CONDITIONAL 95% PAIRED EPISODE-BOOTSTRAP INTERVALS. RATIOS COMPARE MEANS OF TRAINING-SEED ENDPOINTS.

Comparison	Limb	Ratio [95% interval]
0.5 vs. 0 (F)	LA	1.797 [1.645, 1.977]
	RA	1.070 [0.963, 1.167]
	LH	7.636 [4.972, 8.286]
	RH	1.042 [0.954, 1.171]
1 vs. 0.5 (M)	LA	0.983 [0.946, 1.041]
	RA	0.989 [0.920, 1.106]
	LH	0.995 [0.974, 1.012]
	RH	0.939 [0.898, 0.988]

TABLE VII

FULL-STATE $p = 0.5$ VERSUS $p = 0$, SEPARATELY BY TRAINING SEED. RELIANCE IS A DIFFERENCE; ALL OTHER ENTRIES ARE CANDIDATE/BASELINE RATIOS. LA/RA/LH/RH DENOTE LEFT/RIGHT ARM/HAND. CHECKPOINTS REUSE THE SAME TRAINING RUNS.

Stage	Seed	ΔR	RMSE	LA	RA	LH	RH
100	42	0.631	1.121	1.85	1.26	8.21	1.06
100	43	0.630	1.103	1.80	1.04	8.14	0.68
100	44	0.453	1.043	1.34	0.93	4.99	1.24
200	42	0.633	1.099	2.13	1.21	9.26	1.00
200	43	0.632	1.093	1.97	1.09	8.49	0.97
200	44	0.527	1.059	1.43	0.93	5.98	1.17

Constant residuals attain amplification T with an identity Jacobian; alternating residuals can cancel. This per-limb/draw bound uses ordinary coordinates and does not imply multiplicative instability.

Target subtraction. Translation at fixed Q is invertible, proving the conditional-information identity. Clipping or quantization can break it; removing encoder state changes the conditioning.

For a learning counterexample, take orthogonal whitened features q, z , target $a = q + \Delta^*(z)$, $\mathbb{E}[\Delta^* q^\top] = 0$, and $\mathbb{E}[\Delta^* z^\top] = \epsilon B$. With population squared loss and zero initialization, linear absolute and relative predictors have

initial state gradients $-2I$ and 0, respectively, but identical visual gradients. Gradient descent gives both visual weights $[1 - (1 - 2\eta)^k]\epsilon B$. Removing the state-gradient block therefore need not accelerate visual learning. This counterexample concerns squared loss, rather than the experimental generative objective.

Consistent state noise. Since $\tilde{D}_h = A_h - \tilde{Q}$, the command residual equals $\tilde{D}_h - \hat{D}_h$. A common squared metric gives identical target and command errors. Retaining the old target instead shifts an exact-target command by ξ ; zero-mean noise still changes the samplewise objective. The empirical pair measures noise and target recomputation jointly.

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